ME 5309 – Mid-Semester Examination

Answer Booklet

Instructions: Attempt any three questions. Each question carries 10 marks. Show working where appropriate.

Question 1 [10 marks]

Q1(a) [4 marks]

(i) f_x = 16/5 = 3.2 pixels, using focal length 16 mm and pixel size 5 microm. (ii) Sensor width is 6.4 mm, so field width = 400x6.4/16 = 160 mm and scale = 0.125 mm/pixel.

Q1(b) [3 marks]

The smallest resolvable force is 0.122 N.

Q1(c) [3 marks]

$$F = e^(- 2 \times 0.1) = 0.818731$$

I used G = (1-F)/(-2) = -0.0906346. So:

$$x_{k+1} = 0.818731 x_k - 0.0906346 u_k$$

Question 3 [10 marks]

Q3(a) [2 marks]

Finite differencing amplifies encoder noise, making the velocity estimate noisy.

Q3(b) [2 marks]

Tustin uses s = (2/T)(z-1)/(z+1).

Q3(c) [4 marks]

The characteristic polynomial is $s^2+(\ell_1+2)s+(2\ell_1+\ell_2)$. I match it to $s^2+10s+100$ and get $\ell_1=8$, $\ell_2=84$.

Q3(d) [2 marks]

The observer is better and the filter is safer in some cases.

Question 4 [10 marks]

Q4(a) [2 marks]

Poles are -2±2j and Nff=8.

Q4(b) [2 marks]

The nominal controller has unity DC gain, so θss=1 and the tracking error is zero even after the actuator gain changes.

Q4(c) [4 marks]

Q4(d) [2 marks]

Integral action helps eliminate error. A problem is that the controller can become unstable.